Nafis Ahtasum

COMPUTATIONAL MATERIALS RESEARCHER · **RESEARCH AND ANALYSIS INSTITUTE (RAI)**

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RESEARCH PROFILE

Materials informatics researcher combining **physics-guided machine learning, high-throughput screening, DFT simulations, and open-science ecosystem leadership**. Specialized in engineering automated screening and materials-discovery frameworks to predict phase stability, optoelectronic phenotypes, and interfacial reaction kinetics in double perovskites and 2D nanomaterials. Founder of **Resogram** (**1,001 active researchers**) and **RAI**, building distributed cloud-HPC workflows and open scientific infrastructure under resource constraints.

EDUCATION

Bangladesh University of Textiles (BUTEX) — *Dhaka, Bangladesh*

B.Sc. in Textile Engineering (Major in Apparel Engineering) — *Jan. 2019 – Aug. 2026*

- CGPA: 2.88/4.00
- **Undergraduate Thesis:** *Comparative Analysis of Microfiber (Microplastic) Release from Industrial Garment Washing Techniques Using Machine Learning and Optical Image Processing*.
- **Relevant Coursework:** Engineering Mathematics, Computational Modeling, Physical Chemistry, Polymer Science, Fiber Physics, Statistical Quality Control.

PUBLICATIONS & PREPRINTS (Full list on Google Scholar)

- **Ahtasum, N.**, Sohan, S. R., Himel, M. M. A., Hassan, M. Z., Moklis, M. H., and Roni, M. R. A., 2026. **Backward mapping from target optoelectronic phenotypes to chemical genomes for the discovery of lead-free double perovskites**. ***Materials Today Physics*** (Q1, Impact Factor: 9.3), 66, p.102180.
- **Ahtasum, N.**, Sohan, S. R., Himel, M. M. A., Hassan, M. Z., Moklis, M. H., Rashel, M. R., Jamil, H., Islam, A. K., and Tlemcani, M., 2026. **Genome-Guided Interpretable Screening of Phase-Stable, Lead-Free Double Perovskite Absorbers for All-Inorganic Semiconductors, Sensors, and Photovoltaics with DFT-Validated Design Rules**. ***arXiv Preprint***, arXiv:2605.22887.

RESEARCH EXPERIENCE

Computational Researcher, Center for Material, Climate and Energy (CMCE), RAI — **Oct. 2024 – Present**

Supervisor: Dr. M. M. Harussani — *Dhaka, Bangladesh*

- **Title: Backward-Mapped Materials Informatics and High-Recall Metastable Phase Screening**
  Screened 13,088 $A_2BB'X_6$ double perovskites via EA-optimized tree ensembles ($R^2 = 0.93$) and custom high-recall loss functions to preserve synthetically accessible metastable phases ($\Delta E_{\text{hull}} \leq 25$ meV/atom).
- Applied SHAP interpretability to reduce candidate search space by 99.9% and DFT-validated 7 stable endpoints; confirmed 90% reduced excitonic drag for $Cs_2SnGeBr_6$ ($\mu^* = 0.045\, m_0$, $\varepsilon_1(0) = 7.54$).
- **Title: First-Principles Adsorption Dynamics and Stimuli-Responsive Drug Delivery on 2D MXenes**
  Investigated adsorption mechanisms and pH-responsive release kinetics of capecitabine on Ti_3C_2 MXene via DFT and MD simulations; quantified strong exothermic binding ($E_{\text{ads}} \approx -3.2$ eV) stabilized by charge transfer (0.3–$0.4\,|e|$) and demonstrated tumor-acidic release kinetics ($\Delta E_{\text{rel}} \approx +3.2$ eV).
- **Title: High-Throughput DFT Screening of Deep Generative Candidates for HER-Suppression**
  Conducted Density Functional Theory (DFT) calculations on metallic candidates generated via a Deep Generative Framework ($> 5,000$ database compounds, 400k structures), isolating 18 HER-suppressing Co-Zr-X (X = Ba, Ti) trimetallics for cathodic glycerol electroreduction.

Research Collaborator, Department of Computer Science, University of Idaho — **June 2026 – Present**

Supervisor: Prof. Hasan Jamil — *Remote, USA*

- **Title: ML Surrogate-Driven Inverse-Design Pipeline Development for Perovskite Halide and Oxides**
  Developing inverse-design computational workflows, curating benchmark materials datasets, and training ML surrogate models to screen phase-stable perovskite oxide candidates for optoelectronic applications.
- **Title: Genome-Guided Interpretable Screening of Phase-Stable, Lead-Free Double Perovskite Absorbers for All-Inorganic Semiconductors, Sensors, and Photovoltaics with DFT-Validated Design Rules**

Deployed a genome-guided, physics-informed ML framework across 3 descriptor families with staged constraint filtering over $13,000$ compositions, isolating 5 DFT-validated, strictly stable ($E_{\text{hull}} \leq 0$ meV/atom) optoelectronic semiconductors.

Research Collaborator, Universidade de Évora — **Jan. 2025 – Apr. 2026**
Supervisors: Prof. Dr. Mouhaydine Tlemcani and Dr. Masud Rana Rashel — *Remote, Portugal*

- **Title: Accelerated discovery of stable lead-free halide double perovskites for optoelectronics via evolutionary-tuned machine learning and density functional theory**
  Designed end-to-end computational screening workflows, curated materials databases, and developed predictive machine learning pipelines for DFT-validated lead-free double perovskite discovery.

Undergraduate Researcher, Bangladesh University of Textiles (BUTEX) — **Jan. 2024 – May 2025**
Supervisors: Most. Jesmin Nahar Kamona and Nusrat Jahan — *Dhaka, Bangladesh*

- **Title: Experimental Quantification and Microscopic Tracking of Microfiber Shedding Dynamics**
  Led experimental filtration, gravimetric analysis, and material characterization via FE-SEM/EDS, FTIR-ATR, and ImageJ optical microscopy; presented findings at the *International Conference on Textile Science and Engineering 2025*.
- **Title: Hydraulic Hot-Press Formulation and Mechanical Characterization of Recycled Waste Composites**
  Formulated compression-molded composite panels from garment offcuts and polypropylene (PP) matrices via a CARVER hydraulic press; mechanical UTM testing outperformed ASTM D350 OSB/plywood benchmarks (*JFPC*, 2024; *ICONFIB 2024*).

Technical & Computational Skills

Materials Informatics & ML:	Material-discovery architectures, Evolutionary algorithms, Tree ensembles, Neural Networks, Physics-informed ML surrogates, Scikit-learn, Matplotlib, Seaborn, NumPy, Pandas.
First-Principles & Atomistic:	Density Functional Theory (DFT), Molecular Dynamics (MD), Quantum ESPRESSO, Gaussian 09W, Materials Studio, VASP, VESTA;
Experimental Characterization:	FE-SEM/EDS, FTIR-ATR, Optical Microscopy, ImageJ particle tracking, Universal Testing Machine (UTM), Compression molding (CARVER press).
Software & Infrastructure:	Python, Linux/Bash shell scripting, Git/GitHub, LaTeX, OriginPro, Mendeley, Zotero.

Open-Science Platforms & Leadership

2025–Present **Resogram – The Social Network for Researchers** — resogram.com
Founder & Product Lead

- Conceptualized and launched **Resogram**, an academic collaboration platform connecting **1,001 active researchers** globally with **91 research publications/posts**.
- Directed feature architecture for interdisciplinary collaborator matching, open peer review, research dissemination, and conference tracking.

2024–Present **Research and Analysis Institute (RAI)** — ra-institute.org
Founder & Managing Director

- Established an open-access research institute providing structured computational materials science training, HPC access, and research mentorship to early-career scholars.
- Produced global seminar series (*Aspire Talks*, *RAI Research Podcast*) featuring engineering leads from Google, Meta, and Tesla.

Teaching, Supervision & Mentoring

Training Programs & Technical Workshops (Research and Analysis Institute)

Oct. 2025	**Literature Navigation and Critical Analysis**, Scientific Synthesis Methods (*Instructor*; 44 students)
Sep. 2025	**Artificial Intelligence in Scientific Research**, AI/ML Applications in Physics/Materials (*Co-Instructor*; 32 students)
Nov. 2024	**7-Day Hands-on DFT Simulation Training**, First-Principles Calculations & Property Prediction (*Instructor*; 56 students)
Oct. 2024	**Elite Software Training for Research**, Computational Toolchains: Quantum ESPRESSO, Gaussian, Materials Studio (*Instructor*; 122 students)

Research Supervision (RAI Next Generation Researchers Initiative)

2025–2026 **Point Defect Kinetics & Exciton Dynamics in Co-Doped $Cs_2AgBiCl_6$** | *Co-Supervisor (DFT)*
2025–2026 **First-Principles Adsorption Dynamics of H_2SO_4 on MXene Monolayers** | *Co-Supervisor (DFT)*

Additional research Experience

Lead Research Fellow, V-Trion Textile Research GmbH — **July 2025 – Present**
Supervisor: Dr. Ashaduzzaman Khan — *Remote, Austria*

- Directing and mentoring 6 remote research teams (28 undergraduate/graduate students), supervising 6 critical review papers on silicon-anode binders, solid polymer electrolytes, wearable biosensors, wound-infection tracking, biodegradable DAC sorbents, and electrochemical CO_2 conversion pathways.

Science Policy & Systems Researcher, PIRI-BD Initiative — **Apr. 2025 – Present**
Supervisors: Md. Rafiul Alam Roni and Prof. A. K. M. Kamrul Islam — *Dhaka, Bangladesh*

- Co-authoring the strategic vision manuscript *"Towards the Establishment of a Private Interdisciplinary Research Institute"* alongside international faculty and students (NC A&T, Univ. of Iowa, Univ. of Maine, Univ. of Évora), formulating operational blueprints for research governance, funding models, and infrastructure.

Professional Experience

2025-Present	**Researcher**, Center for Material, Climate, and Energy, Research and Analysis Institute (RAI), Dhaka-1208, Bangladesh
2026-Present	**Managing Director**, Research and Analysis Institute (RAI), Dhaka-1208, Bangladesh
2025-2026	**Research Assistant**, UniversaPulsar Lda (Spin-off of University of Évora), Évora, Portugal
2025	**Industrial Trainee**, Knit Concern Ltd., Narayanganj, Bangladesh

Global Outreach & Technical Seminars

2025 **Program Director & Executive Producer**, RAI Global Lecture Series
Organized and hosted featured technical lectures with international scholars and industry leads:
- **MIT Media Lab:** *Innovations in Piezoelectricity & Bioengineering* (Ikra Iftekhar Shuvo, MIT)
- **Industry Keynotes:** Featured speakers from **Google** (Shahed Ahmed), **Meta** (Iqbal Hossain), and **Tesla** (Alan Kabir)
- **European Research Institutes:** Lectures with CNRS France, Univ. of Liège, Univ. of Helsinki, and V-Trion Austria

Academic References

Most. Jesmin Nahar Kamona — *Dhaka, Bangladesh*
Assistant Professor, Department of Apparel Engineering, Bangladesh University of Textiles Email: *jasmine@butex.edu.bd*

Prof. Hasan Jamil, Ph.D. — *Moscow, ID, USA*
Associate Professor, Department of Computer Science, University of Idaho — Email: *jamil@uidaho.edu*

Dr. M. M. Harussani, Ph.D. — *Kuala Lumpur, Malaysia*
Principal Investigator, CMCE, RAI | *Postdoctoral Fellow, Universiti Malaya* — Email: *mmharussani@um.edu.my*